

Step into the World of Code

Syllabus Overview

COURSE INFORMATION:

Course Title:	Python Programming for Beginners
Target Audience:	Students aged 12–16 with little to no prior programming experience.
Level:	Beginner
Duration:	40 Classes, 1 Hour Each
Mode of Delivery:	Online or in-person, hands-on learning.
Prerequisites:	None. The course is designed for absolute beginners.

Teaching Approach:

- ♦ Interactive lessons with step-by-step guidance.
- ♦ Project-based learning to reinforce concepts.
- ♦ Collaboration and peer feedback to enhance understanding.

UNIT 1 INTRODUCTION TO PYTHON

Class 1: Getting Started with Python

Goal: Learn the basics of Python and set up the environment.

Activity: Write a program that displays your name and a fun fact about yourself.

Class 2: Variables and Data Types

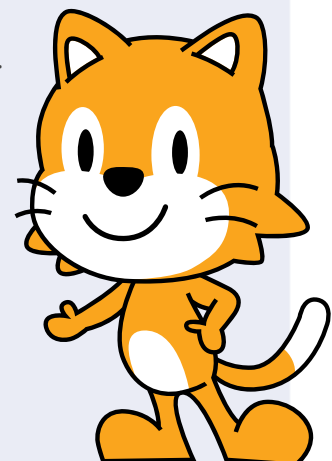
Goal: Understand how to use variables and data types.

Activity: Create a program to calculate the area of a rectangle.

Class 3: Basic Input and Output

Goal: Learn how to take user input and format output.

Activity: Write a program that greets the user and prints their favorite hobby.





UNIT 2 CONTROL STRUCTURES

Class 4: Conditional Statements

Goal: Learn to use if, elif, and else statements.

Activity: Create a program that determines if a user is eligible for a competition based on age.

Class 5: For Loops

Goal: Understand for loops and iteration control.

Activity: Write a program that prints the first 10 multiples of a number entered by the user.

Class 6: While Loops

Goal: Use while loops to repeat actions.

Activity: Create a program that prints the first 10 multiples of a number using while loops.

UNIT 3 FUNCTIONS AND PROBLEM-SOLVING

Class 7: Introduction to Functions

Goal: Learn to write and call functions with parameters and return values.

Activity: Create a function to calculate the factorial of a number.

Class 8: Using Built-In Functions

Goal: Explore common functions like len(), max(), and sum().

Activity: Write a program to find the largest and smallest number in a list.

Class 9: Debugging Basics

Goal: Identify and fix errors in Python programs.

Activity: Debug a pre-written program and make it functional.

UNIT 4 DATA STRUCTURES

Class 10: Lists

Goal: Learn to create, update, and manipulate lists.

Activity: Build a program to manage a to-do list.

Class 11: Dictionaries

Goal: Understand key-value pairs and how to use them.

Activity: Create a dictionary of favorite books and authors.

Class 12: Tuples and Sets

Goal: Explore immutable data structures and unique collections.

Activity: Write a program that identifies unique hobbies entered by the user.

UNIT 5 FILE HANDLING

Class 13: Reading and Writing Text Files

Goal: Learn to read from and write to files in Python.

Activity: Build a program to store and retrieve a list of student names.

Class 14: Error Handling

Goal: Use try, except, and finally blocks to manage errors.

Activity: Create a program that reads a file and handles missing file errors gracefully.

Class 15: Nested Loops and Patterns

Goal: Create patterns using nested loops.

Activity: Write a program that prints a triangle pattern using stars.

UNIT 6 DATA VISUALIZATION BASICS



Class 16: Introduction to Data Visualization

Goal: Use Matplotlib to create simple charts.

Activity: Display monthly rainfall using a bar chart.

Class 17: Customizing Graphs

Goal: Learn to add titles, labels, and legends to charts.

Activity: Visualize a student's weekly study hours.

Class 18: Advanced Graph Types

Goal: Create scatter plots and pie charts.

Activity: Compare time spent on different subjects using a pie chart.

UNIT 7 FINAL PROJECT

Class 19: Planning the Final Project

Goal: Brainstorm and outline the final project.

Activity: Write a plan for a project such as an expense tracker or quiz game.



Class 20: Building and Presenting the Final Project

Goal: Develop and showcase the final project.

Activity: Present projects to peers and celebrate accomplishments.

UNIT 8 SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

Class 21: Introduction to SDLC

Goal: Understand SDLC phases and their application in Python.

Activity: Relate SDLC phases to a sample Python project.

Class 22: Project Planning

Goal: Define objectives, features, and milestones.

Activity: Create a project charter.

Class 23: Gathering Requirements

Goal: Document functional and non-functional requirements.

Activity: Create user stories for the project.

Class 24: Designing the Project

Goal: Create flowcharts and structure the project's codebase.

Activity: Develop diagrams for scalability and functionality.

Class 25: Development Phase - Part 1

Goal: Write the core functionality.

Activity: Modularize code into functions and add basic features.

Class 26: Development Phase - Part 2

Goal: Enhance the project with advanced features.

Activity: Implement additional functionality and test the code.

Class 27: Testing and Debugging

Goal: Test and debug projects systematically.

Activity: Write test cases and debug using Python's unittest module.

Class 28: Refinement and Deployment

Goal: Polish and share the project.

Activity: Finalize visuals, save as an executable, and upload to GitHub.

Class 29: Documentation

Goal: Write maintainable code with documentation.

Activity: Create a README file and add comments to the code.

Class 30: Final Project Presentation

Goal: Showcase SDLC-based projects.

Activity: Reflect on challenges and share learning outcomes.

UNIT 9 ADVANCED PYTHON APPLICATIONS AND MINI-PROJECTS

Class 31: Chatbot Development

Goal: Build a chatbot using string operations and conditional logic.

Activity: Create a chatbot that responds to user inputs.

Class 32: Personal Budget Calculator

Goal: Develop a program for monthly budget calculations.

Activity: Use file handling to save and retrieve budget data.

Class 33: Hangman Game

Goal: Create a text-based guessing game using lists and loops.

Activity: Develop a hangman game where the user guesses letters.

Class 34: Rock, Paper, Scissors Game

Goal: Create an interactive game using randomization and user input.

Activity: Build a rock-paper-scissors game where the user plays against the computer.

Class 35: Quiz Application with Timer

Goal: Create a timed quiz game with lists and dictionaries.

Activity: Develop a quiz where users must answer within a set time limit.

Class 36: Tic-Tac-Toe Game

Goal: Build a two-player text-based strategy game.

Activity: Write a program for a Tic-Tac-Toe game with win conditions.

Class 37: Weather Data Visualization

Goal: Use Matplotlib to graph weather data from a file.

Activity: Read weather data and plot it as a line chart.

Class 38: Student Grades Visualizer

Goal: Calculate and display student grades using charts.

Activity: Create bar charts for grades using dictionaries and loops.

Class 39: Simple Password Manager

Goal: Build a secure password storage tool with basic encryption.

Activity: Write a password manager to store and retrieve passwords.

Class 40: Event Countdown Timer

Goal: Develop a program to track upcoming events.

Activity: Calculate days remaining until a user-specified event.



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